

Clustering / topic modeling

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DIGEORGE SYNDROME; DGS

Alternative titles; symbols

CHROMOSOME 22q11.2 DELETION SYNDROME
HYPOPLASIA OF THYMUS AND PARATHYROIDS
THIRD AND FOURTH PHARYNGEAL POUCH SYNDROME

Other entities represented in this entry:

DIGEORGE SYNDROME CHROMOSOME REGION, INCLUDED; DGCR, INCLUDED

TAKAO VCF SYNDROME, INCLUDED
CATCH22, INCLUDED

Phenotype-Gene Relationships

Location	Phenotype	Phenotype MIM number	Inheritance (in progress)	Phenotype mapping key	Gene/Locus	Gene/Locus MIM number
22q11.21	DiGeorge syndrome	188400	AD	3	TBX1	602054

Clinical Synopsis

TEXT

A number sign (#) is used with this entry because DiGeorge syndrome is caused by a 1.5- to 3.0-Mb hemizygous deletion of chromosome 22q11.2. Haploinsufficiency of the TBX1 gene (602054) in particular is responsible for most of the physical malformations. There is evidence that point mutations in the TBX1 gene can also cause the disorder.

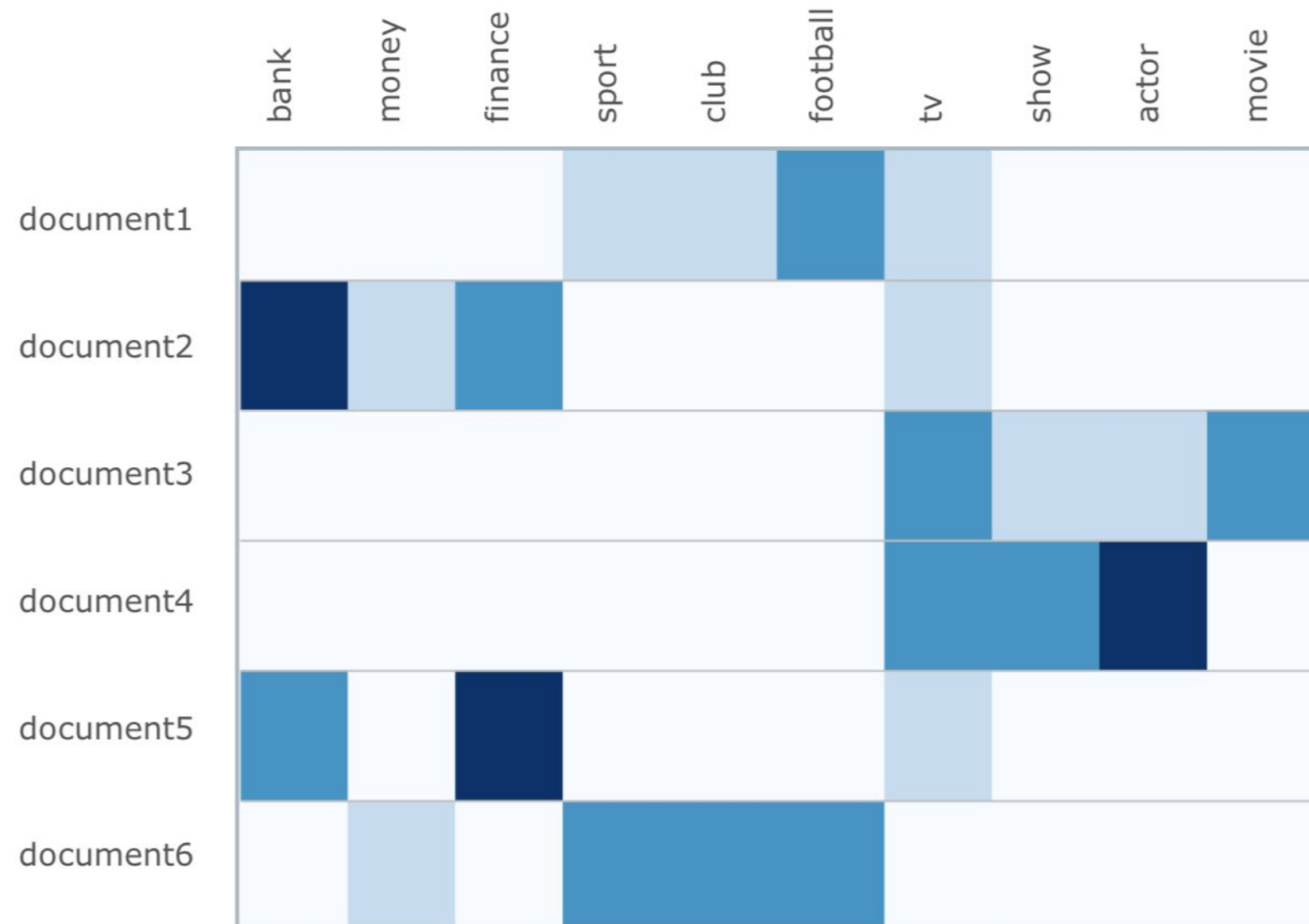
Description

DiGeorge syndrome (DGS) comprises hypocalcemia arising from parathyroid hypoplasia, thymic hypoplasia, and outflow tract defects of the heart. Disturbance of cervical neural crest migration into the derivatives of the pharyngeal arches and pouches can account for the phenotype. Most cases result from a deletion of chromosome 22q11.2 (the DiGeorge syndrome chromosome region, or DGCR). Several genes are lost including the putative transcription factor TUPLE1 which is expressed in the appropriate distribution. This deletion may present with a variety of phenotypes: Shprintzen, or velocardiofacial, syndrome (VCFS; 192430); conotruncal anomaly face (or Takao syndrome); and isolated outflow tract defects of the heart including tetralogy of Fallot, truncus arteriosus, and interrupted aortic arch. A collective acronym CATCH22 has been proposed for these differing presentations. A small number of cases of DGS have defects in other chromosomes, notably 10p13 (see 601362). In the mouse, a transgenic Hox A3 (Hox 1.5) knockout produces a phenotype similar to DGS as do the teratogens retinoic acid and alcohol.

Nomenclature

DiGeorge syndrome overlaps clinically with the disorder described by the Japanese as 'conotruncal anomaly face syndrome' (Kinouchi et al., 1976; Takao et al., 1980; Shimizu et al., 1984), where the cardiovascular presentation is the focus of attention. The term conotruncal anomaly face syndrome is cumbersome and has the disadvantage of using embryologic assumptions as a title. It would be appropriate to use Takao syndrome for those cases with a preponderant cardiac presentation in contrast to the low T cell and hypocalcemic presentation in infancy of DiGeorge syndrome and the craniofacial and palatal abnormalities typical of Shprintzen syndrome. These 3 phenotypes may be seen in the same family and most cases of all 3 categories have been shown to have a 22q11 deletion. This led Wilson et al. (1993) to propose the acronym CATCH22 (Cardiac Abnormality/abnormal facies, T cell deficit due to thymic hypoplasia, Cleft palate, Hypocalcemia due to hypoparathyroidism resulting from 22q11 deletion) as a collective acronym for those with the common genetic etiology. Shprintzen (1994) objected to 'lumping' velocardiofacial syndrome with the DiGeorge anomaly, arguing that there is 'no valid evidence to suggest that velocardiofacial syndrome is etiologically heterogeneous...[whereas] the DiGeorge anomaly is known to be so.' Hall (1993) cited data of Driscoll et al. (1993) indicating that velocardiofacial syndrome is etiologically heterogeneous. She stated that '...68% of Shprintzen syndrome patients...have been recognised to have deletions of 22q11.' Shprintzen (1994) refuted her statement, maintaining that it could accurately be stated that deletion was found in 68% of patients sent to the Driscoll laboratory with a diagnosis of velocardiofacial syndrome made by other clinicians. Shprintzen (1994) said that in his sample, 100% had deletion. 🌱

Document-Term Matrix **A**
(6 rows x 10 columns)



Topics in NMF model (Frobenius norm)

Topic 1

just
don
people
like
think
good
know
time
make
way
really
ve
new
say
want
right
use
ll
did
going

Topic 2

windows
dos
use
using
program
window
os
drivers
application
ms
help
running
software
pc
files
screen
mode
work
card
manager

Topic 3

god
jesus
bible
faith
christian
christ
christians
does
sin
heaven
believe
lord
life
mary
church
atheism
love
human
belief
religion

Topic 4

thanks
know
does
mail
advance
hi
info
interested
anybody
email
card
looking
help
appreciated
like
information
send
video
list
need

Topic 5

00
sale
condition
10
card
today
price
250
asking
offer
new
cd
equipment
tape
15
50
20
car
pay
contact

Topic 6

edu
soon
send
internet
university
mit
com
ftp
cc
mail
article
hope
pub
information
home
blood
mac
program
email
cs

Topic 7

file
problem
files
format
win
sound
pub
read
ftp
save
image
help
available
memory
create
running
site
version
copy
self

Topic 8

game
games
team
win
year
play
season
players
nhl
runs
goal
hockey
defense
player
bad
division
flyers
leafs
toronto
red

Topic 9

drive
drives
hard
disk
floppy
software
mac
mb
controller
scsi
computer
rom
apple
power
internal
problems
problem
access
digital
pc

Topic 10

key
chip
clipper
keys
encryption
government
public
secure
use
enforcement
phone
nsa
communications
encrypted
law
clinton
security
used
message
user

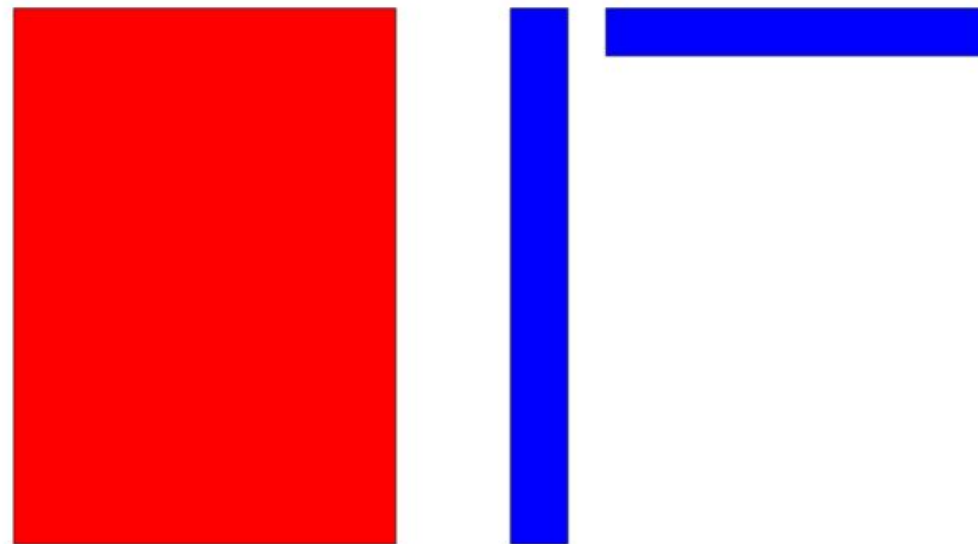
- 1 Aproximaciones Low-Rank para Clustering
- 2 (NMF) Factorización de matrices no-negativas
- 3 NMF
- 4 Aplicación de NMF para extracción de tópicos en texto

Aproximaciones Low-Rank para Clustering

Una matriz X de rango r admite una factorización de la forma

$$X = BC^T, \quad B \in \mathbf{R}^{m \times r}, \quad C \in \mathbf{R}^{n \times r}$$

X es aproximada con bajo rango (low-rank) cuando $\text{rango}(X) \ll \min(m, n)$



(NMF) Factorización de matrices no-negativas

- Grupo de algoritmos de análisis multivariado y álgebra lineal donde una matriz X es factorizada en dos matrices W y H
- Cada columna de X es aproximada por una combinación lineal no-negativa de las columnas de W , donde los coeficientes de mezcla corresponden a las columnas de H
- Las tres matrices tienen elementos no-negativos
- Usado en sistemas recomendadores, procesamiento de audio, agrupamiento de texto.

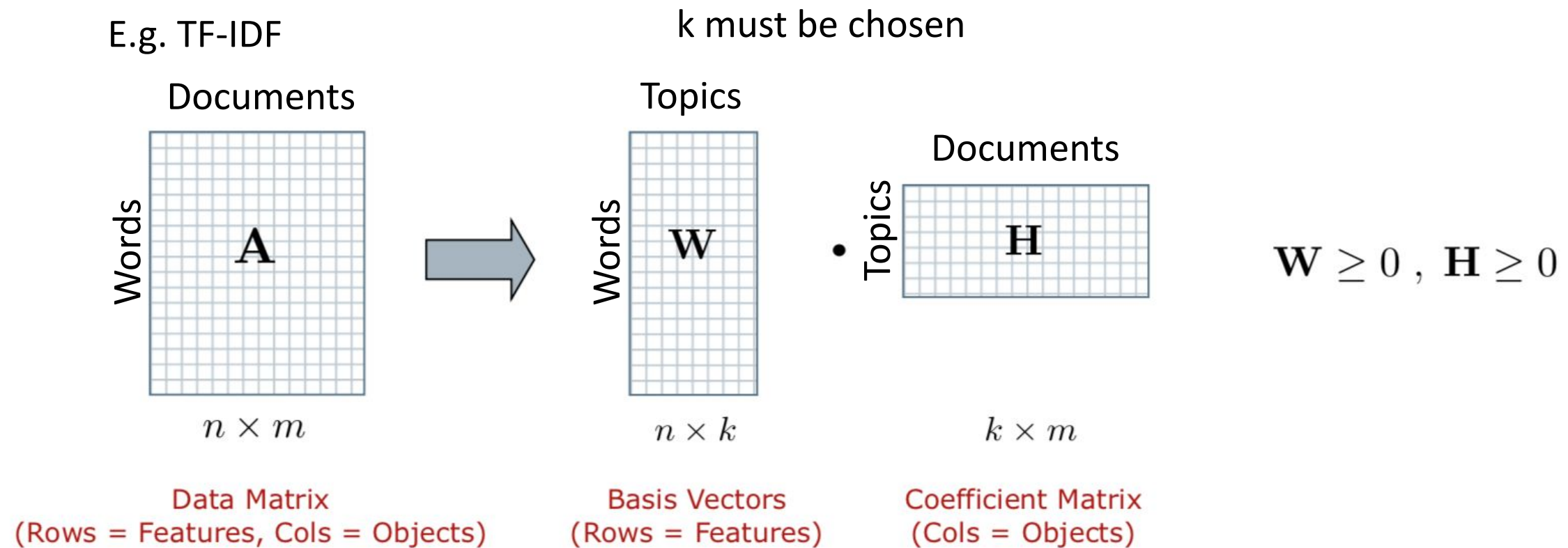
NMF

- Dada una matriz no-negativa $X \in \mathbf{R}^{m \times n}$ y un $k \in \mathbf{Z} \ll \min(m, n)$
- Encuentra matrices no-negativas $W \in \mathbf{R}^{m \times k}$ y $H \in \mathbf{R}^{k \times n}$ tales que minimizan

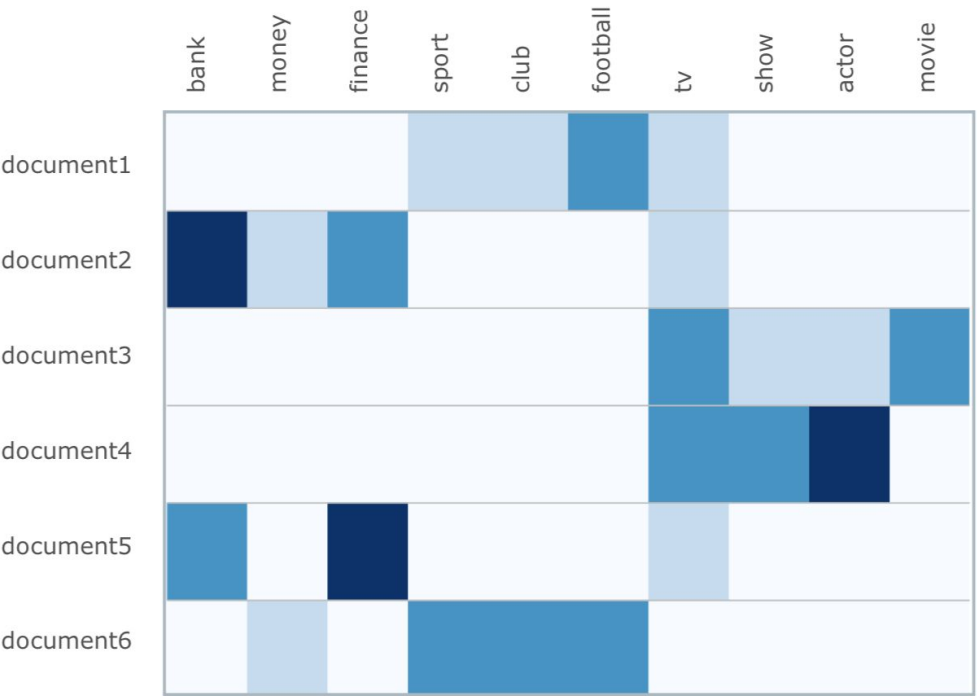
$$\|X - WH\|_F^2 = \sum_i \sum_j (X_{ij} - [WH]_{ij})^2$$

* W : base para un espacio k -dimensional, la i -ésima columna de H : corresponde a representación k -dim de i -ésima columna de X

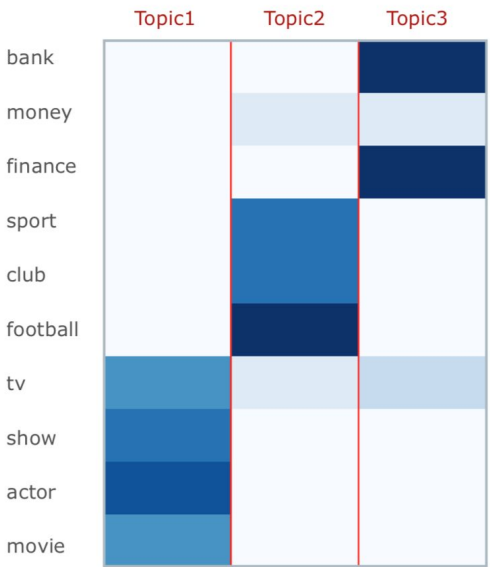
Matrix factorization for topic models



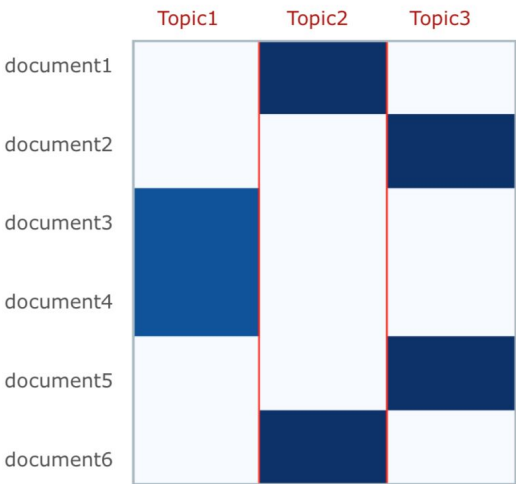
Document-Term Matrix **A**
(6 rows x 10 columns)



Basis vectors **W**: topics
(clusters)



Coefficients **H**: memberships
for documents



Matrix factorization for topic models

Inputs

Non-negative data matrix (**A**), number of basis vectors (k), initial values for factors **W** and **H** (e.g. random matrices).

Optimization

Euclidean
Distance
(Lee & Seung, 1999)

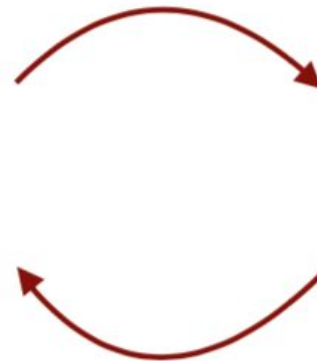
$$\frac{1}{2} ||\mathbf{A} - \mathbf{WH}||_F^2 = \sum_{i=1}^n \sum_{j=1}^m (A_{ij} - (WH)_{ij})^2$$

1. Update **H**

$$H_{cj} \leftarrow H_{cj} \frac{(W\mathbf{A})_{cj}}{(W\mathbf{WH})_{cj}}$$

2. Update **W**

$$W_{ic} \leftarrow W_{ic} \frac{(\mathbf{AH})_{ic}}{(\mathbf{WHH})_{ic}}$$



Topics of OMIM

Topic #0: nose nasal bridge palpebral fissures face philtrum hypertelorism ears forehead folds epicanthal short downslanting flat micrognathia eyes head arched tip	anteverted hands mouth ptosis nares depressed midface upper eyebrows lips height hair upslanting feet narrow clinodactyly stature growth bossing skeletal	Topic #1: muscle tissue soft weakness creatine kinase biopsy proximal serum muscular myopathy atrophy onset changes myopathic emg fibers muscles fiber progressive girdle dystrophic contractures hypotonia hypertrophy	Topic #3: ribs spine clavicles sternum scapulae height pelvis chest short stature growth vertebral skeletal bones femoral platyspondyly epiphyses iliac scoliosis coxa	metaphyses metaphyseal skull shaped pectus bone narrow lumbar limbs bodies vara genu ossification dysplasia hands wings vertebrae dwarfism kyphosis epiphyseal	Topic #39: bronchiectasis sinusitis respiratory situs otitis inversus chronic dynein ciliary lung bronchitis airways clearance infections rhinitis outer cilia arms defective	microscopy electron immotile oxide nitric nasal inner motility lungs infertility cells sperm nose abdomen ears dextrocardia early polyps onset infancy pneumonia
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Ejemplo de las palabras más representativas en11tópicos

Topic 1	Topic 2	Topic 3	Topic 4	Topic 5	Topic 6	Topic 7	Topic 8	Topic 9	Topic 10	Topic 11
<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
internacionalización	región	desarrollo	importante	prestigio	problemas	mejoras	tiempo	inclusiva	formación	académicos
nacional	valparaíso	medio	áreas	futuro	sociales	universidades	profesores	innovadora	profesional	comunidad
investigación	institución	vinculación	vanguardia	educación	desarrollo	aporte	profesional	estudiantes	calidad	estudiantes
región	comunidad	investigación	personas	estudiantes	temas	país	cambio	aprendizaje	nuevas_tecnologías	personas
liderazgo	entorno	estudiantes	estudios	programas	excelente	calidad	mejoras	abierta	procesos	funcionarios
reconocida	compromiso	sostenible	compleja	calidad	institución	desarrollo	mundo	personas	valores	conocimientos
referente	local	innovadora	espacios	carreras	resolver	conocimientos	tiempos	desarrollar	continua	oportunidad
proyectos	tradición	ambiente	nuevas_tecnologías	chile	público	infraestructura	puedan	investigación	institución	preocupada
áreas	ciudad	permitan	carreras	mejoras	país	enseñanza	gestión	más_inclusiva	trabajo	espacios
manteniendo	nacional	institución	territorio	mundo	comprometida	tres	investigación	calidad	estudiantes	servicio

Asociación texto y tópico

